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ORIGINAL ARTICLE

Explainable ensemble deep learning-based model for brain tumor detection and classification

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Abstract

Brain tumors are very dangerous as they cause death. A lot of people die every year because of brain tumors. Therefore, accurate classification and detection in the early stages can help in recovery. Various deep learning techniques have achieved good results in brain tumor classification. The traditional deep learning methods and training the neural network from scratch are time-consuming and can last for weeks of training. Therefore, in this work, we proposed an ensemble approach depending on transfer learning that utilizes pre-trained models of DenseNet121 and InceptionV3 to detect three forms of brain tumors: meningioma, glioma, and pituitary. While developing the ensemble model, some changes were made to the architecture of pre-trained models by replacing their classifiers (fully connected and SoftMax layers) with a new classifier to adopt the recent task. In addition, gradient-weighted class activation maps (Grad-CAM) are an explainable model to verify results and achieve high confidence. The suggested model was validated using a publicly available dataset and achieved 99.02% accuracy, 98.75% precision, 98.98% recall, and a 98.86% F1 score. The suggested approach outperformed others in detecting and classifying brain tumor MRI data, and verifying results using the explainable model achieved a high degree of trust.

Keywords Explainable AI · Deep learning · Brain tumor · Transfer learning · MRI · CNN

Abbreviations

MRI	Magnetic resonance imaging
CE	Contrast enhanced
AI	Artificial intelligence
CNN	Convolutional neural network
ML	Machine learning
DL	Deep learning
Grad-CAM	Gradient-weighted class activation mapping
ROC	Receiver operating characteristic
AUC	Area under the ROC curve
ROI	Region of interest

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1 Introduction

Brain tumors are among the most severe types of malignant tumors. These are abnormal tissue masses where cells spread and replicate excessively, appearing unregulated by the processes that manage usual cells. Tumors of the brain can cause either focal or generalized neurologic symptoms [1]. There are two types of brain cancers: primary tumors and secondary tumors. Primary brain cancers arise in the brain and may grow in any part of the brain or spinal cord. Primary cancers can be malignant or benign. On the other hand, secondary cancers form within different organs, such as the lungs, breast, and kidneys, before spreading to the brain.

The most frequent forms of brain tumors are meningioma, glioma, and pituitary. Most commonly, gliomas are brain tumors, though they can also occasionally be discovered in the spinal cord. Gliomas are the predominant type of brain cancer found in adults [2]. These cancers are caused by the glial cells, which wrap and support neurons. Meningioma originates in the meninges, the outer three-level tissues that envelop the cerebral area immediately beneath the skull. Meningiomas represent 53.9% of non-malignant cancers, according to United States Central Brain Tumor Registry statistics from 2013 to 2017 [3]. Tumors of the pituitaries are abnormal pituitary gland evolution. The size of this gland is comparable to that of a pea. It is placed behind the nose, near the base of the brain. Some tumors cause the pituitary gland to produce excessive amounts of certain hormones that control critical biological functions.

A range of medical imaging modalities can be used to diagnose brain tumors. Three of the most extensively used technologies are ultrasonography (US), magnetic resonance imaging (MRI), and computed tomography (CT). Magnetic resonance imaging (MRI) is the most frequently used harmless imaging technology. Unlike X-ray and CT imaging, it does not release dangerous ionizing radiation and provides a clearer image. Furthermore, MR allows for more freedom in imaging settings. Two kinds of imaging sequences are frequently utilized. T1-weighted scans provide exceptional structural details; T2-weighted scans, on the other hand, can distinguish between healthy and diseased tissue [4].

In the early stages, accurate diagnosis and classification of various types of cancer are critical. However, this task is challenging because tumors differ in dimension, form, and place. Many machine learning (ML) and deep learning (DL) algorithms are applied for brain tumor classification. ML approaches are time-consuming since they rely on hand-crafted recognition of features for training before categorization. Several studies employed ML approaches to analyze medical images, such as support vector machines [5], k-nearest neighbors [6], and decision trees [7]. While DL techniques can extract features automatically and have proven that they can be used to analyze images in healthcare efficiently, the convolutional neural network (CNN) is one of the most well-known models, as it can distinguish both general and specialized features in medical images. Various studies used DL methods [8–10].

Artificial intelligence is used in many fields, including the medical domain. DL models are complicated and do not interpret the obtained results; hence, increasing confidence and trust in such models in the medical domain should be implemented to remove the gap between AI models and human understanding. Such models that are not interpretable are known as “black boxes” [11]. Explainable AI refers to methods and strategies for implementing models that interpret and explain the produced results. In CNN, the input image can be explained by obtaining weighted gradients of the classification result for the final convolutional layer, which allows us to identify the significant parts with the highest impact on the predicted class, which is then converted into a heat map overlay along with the input image. Various studies have implemented DL with explainable AI in the medical field [12, 13].

DL has proved efficient in many fields, especially in the medical field, which has increased recent research to improve the performance of computer-aided diagnosis systems. For example, Ghafourian et al. [14] presented an ensemble model for brain tumor classification. Firstly, they used the social spider optimization (SSO) segmentation algorithm. Secondly, they employed an ensemble approach comprised of three classifiers to diagnose brain tumors. The three classifiers used were Naive Bayes (NB), support vector machine (SVM), and k-nearest neighbor (KNN). Their proposed method was evaluated on two datasets, BRATS2014 [15] and BR35H [16], and achieved an acceptable accuracy of 98.61% and 99.13%, respectively.

Kang et al. [17] integrated an ensemble model based on transfer learning to extract the deep features by choosing and combining the most significant features extracted from brain MR images. They also utilized nine ML classifiers to differentiate between brain tumor kinds properly. Ensemble feature extraction improved the performance and addressed the problem of one feature from a single model, which may have bad performance. Reza et al. [18] proposed a similar architecture to VGG-16 using dropout regularization to overcome the overfitting problem. In their research work, they found that 20% is the best drop rate to be used. They achieved an accuracy of 99.5% on a combined two publicly available Kaggle datasets [19, 20].

Verma et al. [21] proposed a transfer learning scheme with a configuration of random initialization and freezing convolutional layers. They explored many pre-trained models with different configurations and found that DenseNet201, with the first 5 layers frozen, is the best model in terms of accuracy. Their proposed approach has acquired an accuracy of 98.22% on the Figshare dataset [20]. Gómez-Guzmán et al. [22] identified a model using transfer learning techniques to utilize the InceptionV3 pre-trained network. They also used data augmentation approaches to expand the size of samples in the dataset and overcome the issue of an imbalanced distribution. Their approach was tested on a combined dataset of Figshare [20], BR35H [16], and SARTAJ [23] and gained an accuracy of 97.12%.

Athisayamani et al. [24] utilized the integrated adaptive canny mayfly algorithm (ACMA) to identify the edges of

the brain images in the segmentation process. They also used the spatial gray level dependence matrix (SGLDM) to extract the features and the chimp optimization technique for feature selection. In the final step, the ResNet-152 pre-trained network is used for classification. On the Figshare dataset [20], their proposed technique attained an accuracy of 98.85%. Ayadi et al. [25] proposed an approach for learning the tiny texture of tumors in brain images. The CNN architecture of 3×3 kernels for all convolutional layers with a relatively small stride was developed. Their research has accomplished an acceptable accuracy of 94.74% on the Figshare dataset [20].

Kumar et al. [26] proposed the ResNet-50 with a global average pooling (RNGAP) model using transfer learning. They also employed data augmentation approaches to enhance generalization and resolve overfitting. Their proposed method achieved 97.48% and 97.08% accuracy with and without data augmentation on the Figshare dataset [20]. Alanazi et al. [27] introduced an isolated CNN that underwent training on a two-class dataset. After training, they reutilized the isolated CNN as a pre-trained two-class network through transfer learning techniques to readjust weights to categorize the tumors into subclasses (meningioma, glioma, and pituitary). The researchers of this work found that the best number of layers of isolated CNN is 22 layers, and they obtained an accuracy of 99.33%, 92.67%, and 95.75% on three datasets: BR35H [16], SARTAJ [23], and Figshare [20], respectively. Amran et al. [28] integrated a hybrid DL transfer learning model by replacing the last five layers of Google Net with 14 additional layers of the CNN model. They also used Leaky ReLU [29] to solve the issue of dying neurons caused by ReLU. Their algorithm had an accuracy of 99.20% on the BR35H dataset [16].

Much research has deployed AI models to detect brain tumors using different approaches. Some used single models with transfer learning, others used custom CNN, and others used hybrid models. These approaches have some limitations; hybrid or ensemble models can get features or predictions from more than one model, while stand-alone algorithms using transfer learning rely only on the extracted features from one model. Custom CNN models require more time to train the network from scratch and more training instances to achieve robust models. Using hybrid or ensemble models is a better option because they do not rely on single-model features or predictions as a single model does. Nevertheless, one last point: These models must provide interpretability for their results, especially in the medical field, where misdiagnoses may lead to undesirable effects and delay the treatment period.

DL using ensemble learning and explainable AI motivated us to develop a strategy for improving existing algorithms and providing interpretability that may be utilized for diagnosing various forms of malignant brain tumors. The suggested method used ensemble and transfer learning methods for detecting three main kinds of brain tumors: meningioma, glioma, and pituitary, which were trained and evaluated on a dataset that is free to access on Figshare. The proposed approach used pre-trained models (DenseNet121 [30], InceptionV3 [31]) adopted by removing their classifiers and introducing a recent one for the specific task. In addition, gradient-weighted class activation maps (Grad-CAM) [32] were used to find the significant parts that impact the predicted class most. The suggested approach outperformed previous studies, and applying the explainable methods to the ensemble model enhanced interpretability and increased trust.

The following are contributions to this study paper:

- An explainable model (Grad-CAM) was deployed to the ensemble model to verify the results by localizing the affected areas. Grad-CAM was modified to explain any convolutional layer or a series of layers.
- We employed an ensemble model of DenseNet121 and InceptionV3 pre-trained models using transfer learning techniques to detect and classify three kinds of brain tumors (meningioma, glioma, and pituitary). This model outperforms existing approaches and achieves the highest accuracy on a publicly available dataset.
- The suggested method's findings were obtained using two evaluation methods: Train–test split and stratified k-fold cross-validation to ensure the reliability of the model's performance.
- The proposed model was compared to MobileNet, Xception, ResNet-50, DenseNet121, and InceptionV3, outperforming all of them.
- Compare the suggested strategy with existing approaches.

This research article is organized as stated: Sect. 2 provides an overview of the dataset and the steps used for preprocessing. Section 3 gives a comprehensive explanation of the suggested methodology. Section 4 summarizes the findings of the study. Section 5 contains the discussion. Section 6 gives the conclusion.

2 Dataset and preprocessing

This section describes the dataset and outlines the necessary preparation methods. In this work, we used the contrast-enhanced MRI dataset initially used in [33, 34] and publicly available on Figshare [20], which includes 3064 T1-weighted contrast-enhanced images obtained from Nanfang Hospital, Guangzhou, China, and General Hospital, Tianjin Medical University, China, from 2005 to 2010. The dataset included three kinds of tumors:

meningioma (708 slices), glioma (1426 slices), and pituitary tumors (930 slices) obtained from 233 patients. The image size is 512 × 512 with a pixel dimension of 0.49 × 0.49 mm^2, the thickness of a slice is 6 mm, and the slicing gap is 1 mm. Three experienced radiologists manually delineated the tumor border. The images are stored in .mat files acquired from three planes (axial, coronal, and sagittal). Figure 1 and Table 1 introduce the details of distinct image views of the CE-MRI dataset.

The images were normalized (from .mat to .png). Also, the unnecessary black parts of the images were removed to extract the region of interest (ROI) by applying a threshold to the images and getting the four extreme points of the edges. These extreme points served as the basis for cropping. Finally, they were resized back to their original size of 512 × 512. Figure 2 shows the preprocessing steps for extracting ROI. Due to computational processing restrictions, the MRI scans were downsized to 256 × 256 pixels. Additionally, the dataset was divided into 70% for training and 30% for testing, distributed as follows: 2146 images (496 meningiomas, 999 gliomas, and 651 pituitaries) and 918 images (212 meningiomas, 427 gliomas, and 279 pituitaries), respectively.

Table 1 Details of three distinct planes of MR images

Tumor class	Images	MRI view	No. of MRI images
Meningioma	708	Axial	209
		Coronal	268
		Sagittal	231
Glioma	930	Axial	291
		Coronal	319
		Sagittal	320
Pituitary	1426	Axial	494
		Coronal	437
		Sagittal	495
Total	3064		3064

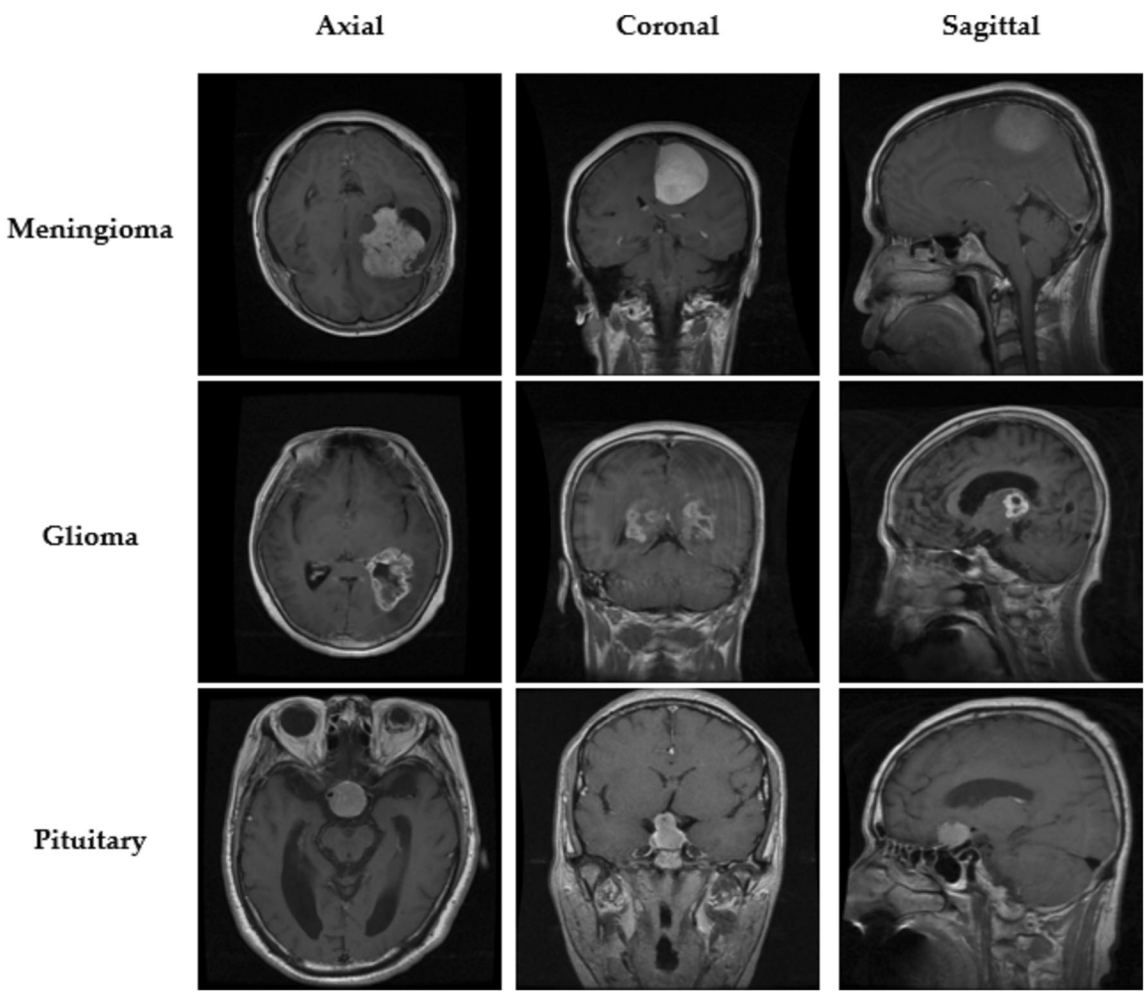

Fig. 1 MR images from three distinct planes of three cancers (meningioma, glioma, and pituitary)

Fig. 2 Preprocessing steps to separate ROI. **a** Input image. **b** Thresholding image. **c** Cropped and resized image

3 Proposed methodology

This section introduces pre-trained networks, explains the suggested methodology and the specific hyperparameters employed, and emphasizes the explainable method.

3.1 DenseNet121

DenseNet121 [30] is an image classification convolutional neural network (CNN) that evolved from earlier CNN architectures like VGG and ResNet. It enhanced the transfer of information and gradients across the network by introducing different connectivity patterns so that each layer directly connects to all subsequent layers. It consists of multiple dense blocks comprising convolutional layers of (1x1 and 3x3) filters. To decrease the size of input feature maps, bottleneck layers (1x1 convolutional layer) were used before each 3x3 convolution, which helped boost computational performance. Moreover, it uses transition layers composed of a 1x1 convolutional layer and a 2x2 average pooling layer to control the growth of feature maps and the number of parameters.

3.2 InceptionV3

InceptionV3 [31] is a convolutional neural network designed for image classification purposes and has proven its efficiency in detecting complex image patterns. It uses inception modules in which each module uses multiple convolutional layers in parallel with varying kernel sizes (1x1, 3x3, 5x5) and max pooling operations. Also, it employed the idea of factorization by using a combination of small convolutions (3x3, 1x7, and 7x1), which helped reduce the number of parameters and computation processing instead of using larger convolution filters. In addition to combating the vanishing gradient problem, it uses auxiliary classifiers at intermediate layers of the network to push proper gradients to the lower layers, which makes them immediately helpful and improves convergence.

3.3 Ensemble model

We proposed an ensemble model of DenseNet121 and InceptionV3 based on transfer learning. The classifiers (fully connected and SoftMax layers) were removed, and a new one was introduced to be trained on the CE-MRI dataset. After training the classifier for both pre-trained networks, both models were combined using the soft voting approach to get the predicted tumor class by computing the mean probability of each class. The predicted class has the highest mean probability. Figure 3 displays the diagram of the suggested methodology.

To avoid overfitting, adopt a dropout layer [35] and L2 regularization [36]. The dropout layer can reduce overfitting by randomly dropping a specified number of neurons using a drop rate while training the network. On the other hand, L2 regularization can help overcome overfitting by applying a penalty term to the loss function, limiting the weights to be small. Table 2 shows the added layers with their parameters for both networks. Due to the limited size of the dataset and the imbalanced classes, data augmentation techniques were applied to overcome this problem. The TensorFlow data augmentation API was used to expand the number of instances and make the model robust against the unseen test data. Table 3 shows the data augmentation parameters. Algorithm 1 gives a further explanation of the proposed method.

Algorithm 1 Ensemble Brain Tumor Classification Using Transfer Learning

```
Input: CE-MRI Dataset, Epochs: 25, Batch size: 10, Learning rate: 0.0001
Step 1: Apply Preprocessing to extract ROI.
Step 2: Apply data augmentation using random transformations with the specified ranges.
For the model in pre-trained models {DenseNet121, InceptionV3}
    Step 3: Load the model with ImageNet weights.
    Step 4: Remove the model classifier.
    Step 5: Add the new classifier layers to the model to be trained on the CE-MRI Dataset.
    For epochs 1 to N, do
        Step 6: Train classifier layers on the CE-MRI Dataset.
        Step 7: Optimizing classifier layers training parameters using Adam optimizer.
        Step 8: Evaluate the model using validation accuracy.
        If there have been no improvements in the past three epochs, do
            Step 9: Reduce the learning rate by half.
        end if
    end for
end for
Step 10: Create an ensemble model by obtaining predictions from both models using a soft voting rule.
Output: Preprocessed MR images
```

3.4 Hyperparameters

Each network was trained with the same parameters illustrated in Table 4 for a constant number of epochs while monitoring the improvement of training and validation accuracy. Also, an adaptive moment estimation (Adam) optimizer [37] was used with a mini-batch size of 10 and an initial learning rate of 0.0001 to train each network for 25 epochs. The learning rate decreased by half when the validation accuracy failed to improve in 3 epochs using a reduced learning rate on plateau callback. The optimal values of hyperparameters were chosen based on some experiments with the models. We trained the models using various values for each parameter while calculating the overall accuracy to select the optimal values and monitor the impact on the models' performance. Figure 4 demonstrates the performance of the models using different hyperparameter values.

3.5 Explainable model

The proposed research employed the gradient class activation map (Grad-CAM) as the explainable model. Grad-CAM is a method of differentiating locations that visualizes why a particular area is chosen or assigned. It can be used for any CNN without further retraining or changing its architecture. Grad-CAM is implemented on the last convolutional layer of DenseNet121 and InceptionV3, as convolutional layers store spatial information missed in the fully connected layer. Additionally, the final convolutional layers have the most significant benefit of combining high-level semantics and spatial data [32]. Grad-CAM can find considerable parts of the input image by obtaining importance weights while backpropagating gradients concerning activations. Furthermore, some modifications were made to Grad-CAM to explain any convolutional layer.

4 Results

This section evaluates the proposed method using various metrics, presents its explainable model results, discusses its computing complexity, and compares it with recent studies. When designing a computer-aided diagnosis system, it is essential to evaluate its performance using a variety of metrics. The proposed model was assessed on the CE-MRI dataset, split into 70% for training and 30% for testing. We obtained the outcomes using the TensorFlow library on a graphics processing unit (GPU) with computer resources, specifically a Nivida RTX 3050 and 16 GB of RAM. The confusion matrix is a standard method for assessing performance in classification problems. It analyzes the classification model in a table of actual and predicted samples from every class. The confusion matrix calculates the most important measures: accuracy, precision, recall, and the F1-score. As stated in [38], the classification metrics for multi-class problems can be derived from binary classification metrics.

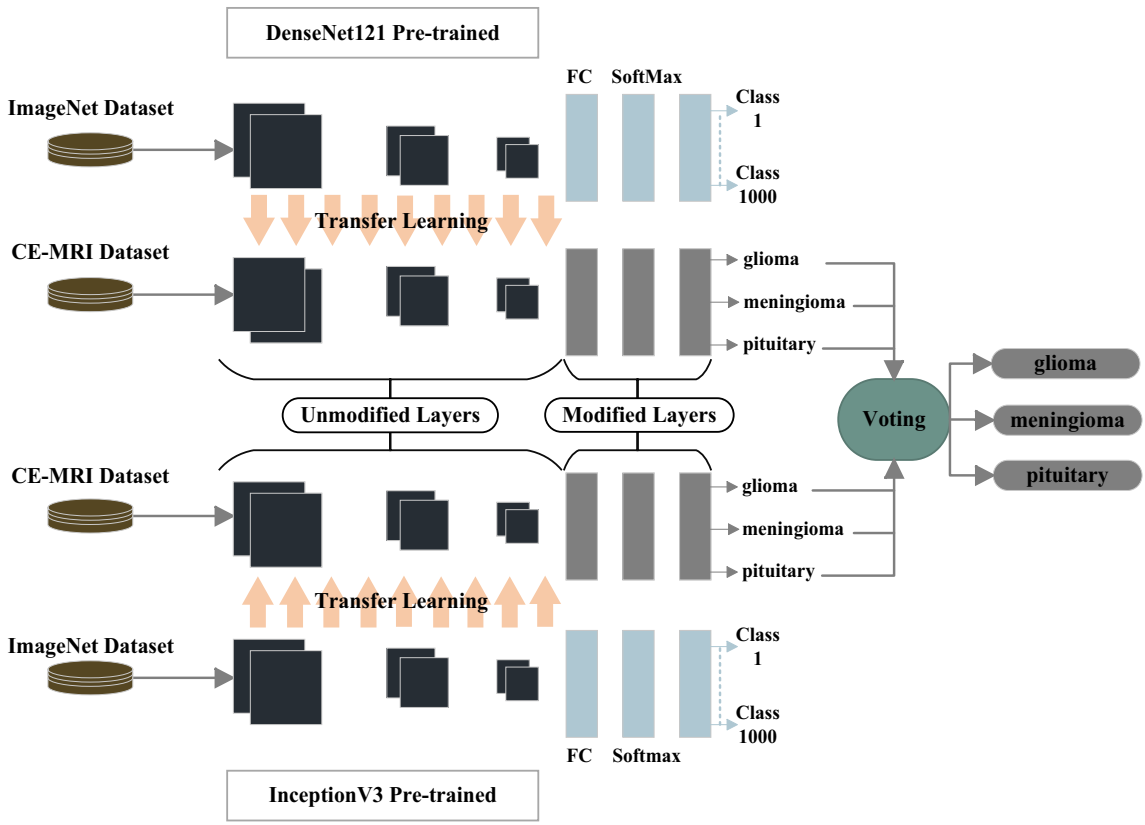

Fig. 3 Diagram of the ensemble model based on transfer learning of DenseNet121 and InceptionV3

Table 2 Details of the classifier added layers

No	Layer	Units/drop rate	Activation	L2 penalty
1	Dropout	0.2	–	–
2	Dense	512	ReLU	0.001
3	Dropout	0.2	–	–
4	Dense	256	ReLU	0.001
5	Dropout	0.2	–	–
6	Dense	3	SoftMax	–

Table 3 Data augmentation parameter values

Augmentation setting	Value
Rotation range	20
Width shift range	0.2
Height shift range	0.2
Shear range	0.2
Zoom range	0.2
Horizontal flip	True

Table 4 Parameter values used in the training process

Parameter	Value
Optimizer	Adam
Epochs	25
Batch size	10
Initial learning rate	0.0001

4.1 Evaluation metrics

Accuracy is the ratio of correctly anticipated observations to all observations. Equation (1) provides the formula for computing accuracy:

$$\text{Accuracy} = \frac{\text{correctly anticipated observations}}{\text{all observations}} \tag{1}$$

Macro-Average Precision the average of each class i precision, where the precision of each class is computed as the ratio of correctly anticipated observations of the positive class i to the labeled positive observations of that class. Equation (2) provides the formula for computing macro-average precision:

$$\text{Precision}_M = \frac{\sum_{i=1}^{N} \frac{TP_i}{TP_i + FP_i}}{N} \tag{2}$$

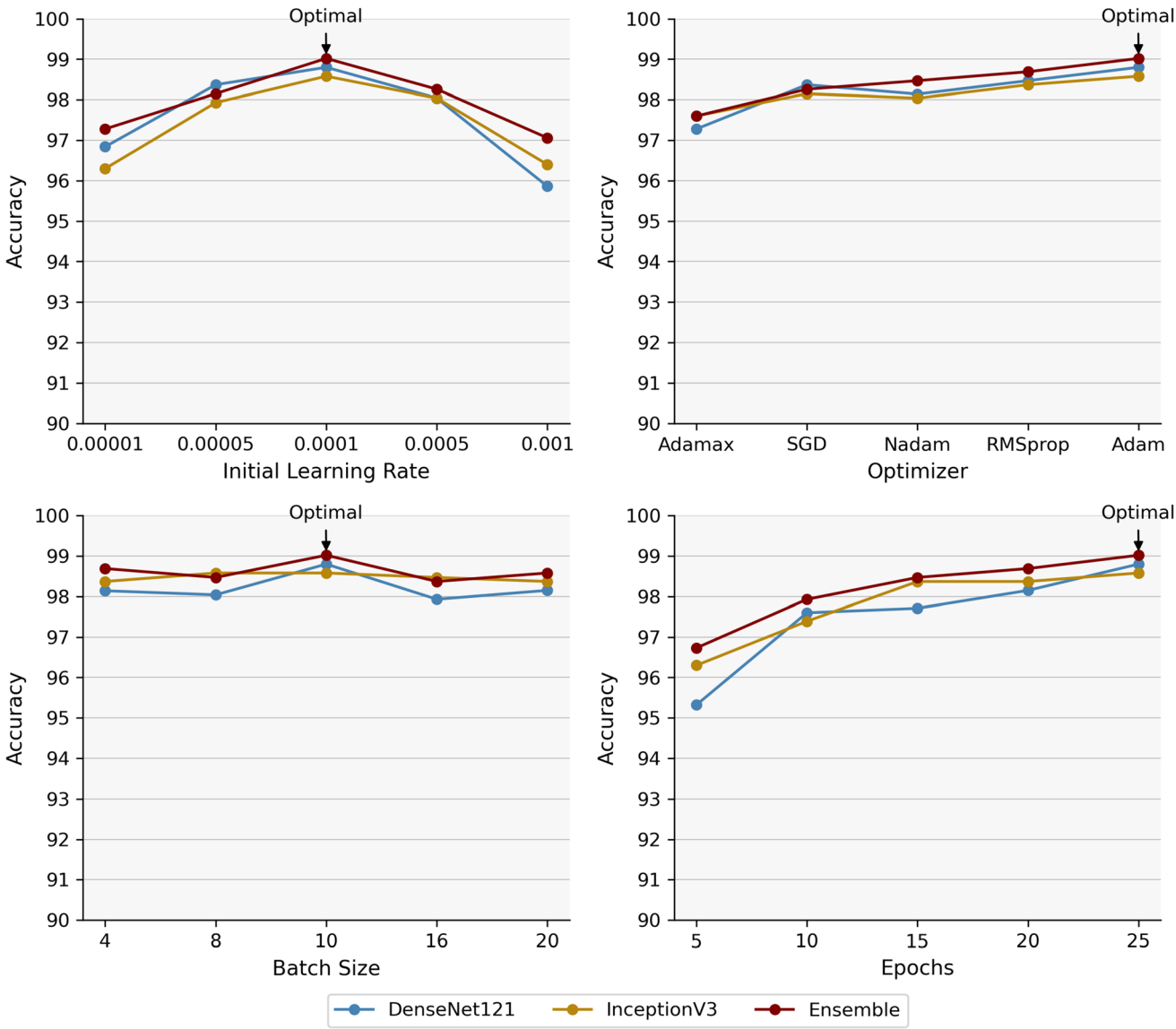

Fig. 4 Impact of different hyperparameter values on the performance of the models

Macro-Average Recall the average of each class i recall, where the recall of each class is computed as the ratio of correctly anticipated observations of the positive class i to the entire positive class i observations. Equation (3) provides the formula for computing macro-average recall:

$$\text{Recall}_M = \frac{\sum_{i=1}^{N} \frac{TP_i}{TP_i + FN_i}}{N} \tag{3}$$

Macro-Average F1-score the average of each class i F1-score, where the F1-score of each class is computed as the mean of the precision and recall values of class i. Equation (4) provides the formula for computing the macro-average F1 score.

$$F1 - \text{score}_M = \frac{\sum_{i=1}^{N} \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}}{N} \tag{4}$$

TP_i is true positive, where the predicted observations were correctly labeled as the positive class i.

FP_i is false positive, where the predicted observations were mislabeled as the positive class i.

TN_i is true negative, where the predicted observations were correctly labeled as the negative class i.

FN_i is false negative, where the predicted observations were mislabeled as the negative class i.

N denotes the number of classes.

Figure 5 represents the training and validation accuracy of DenseNet121 and InceptionV3. Table 5 and Fig. 6 demonstrate the ensemble model's accuracy, precision,

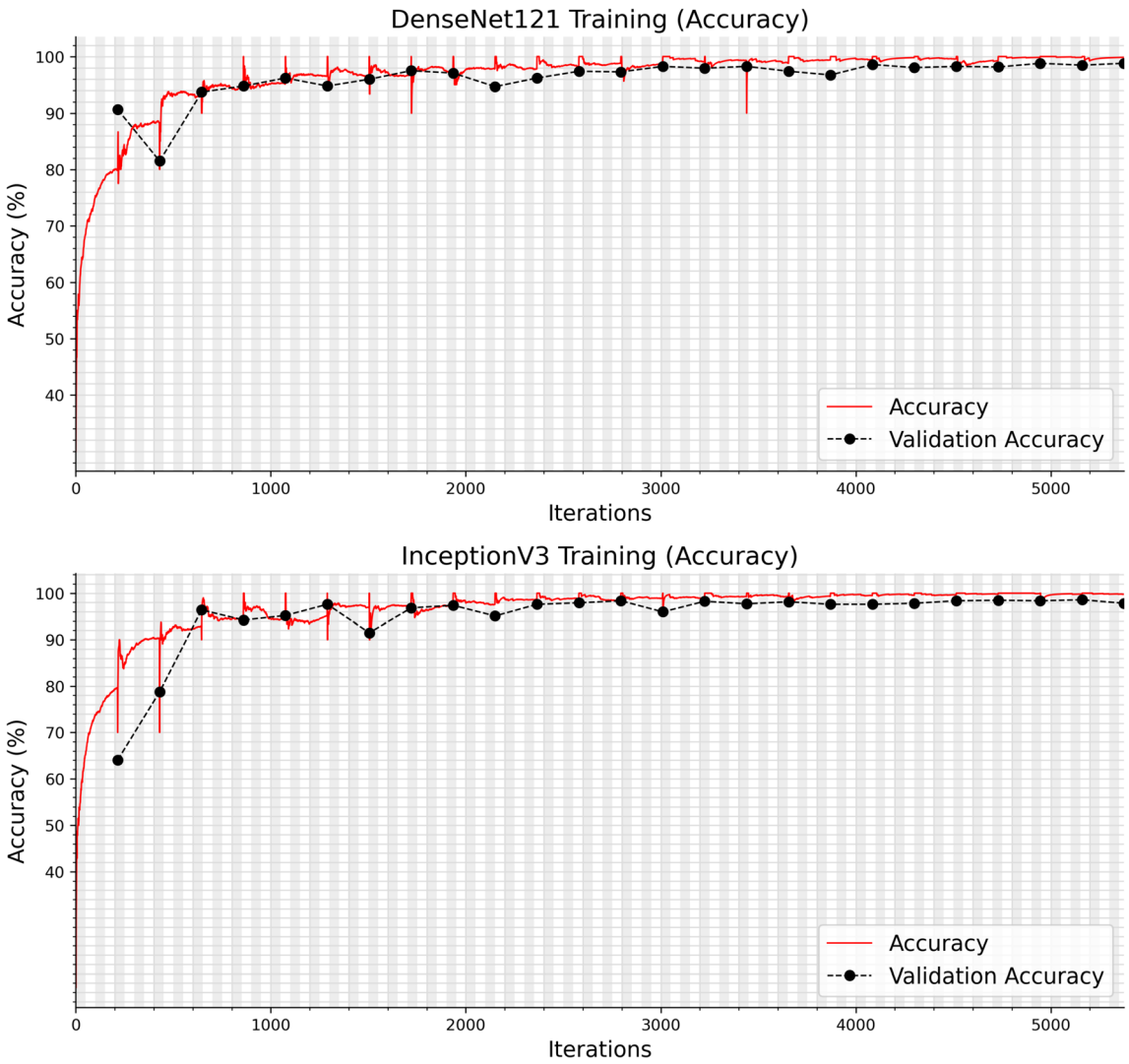

Fig. 5 Training-validation accuracy curve of ensemble model

Table 5 Evaluation metrics for the ensemble model compared with pre-trained networks

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
MobileNet	98.04	97.57	98.00	97.77
Xception	98.15	97.69	97.96	97.82
ResNet-50	98.47	98.06	98.39	98.22
DenseNet121	98.80	98.48	98.74	98.61
InceptionV3	98.58	98.26	98.59	98.42
Ensemble (DenseNet121 + InceptionV3)	99.02	98.75	98.98	98.86

recall, and F1-score evaluation measures compared with the pre-trained networks MobileNet, Xception, ResNet-50, DenseNet121, and InceptionV3. Figure 7 illustrates the confusion matrix, highlighting the corrected and uncorrected labeled observations by each pre-trained and ensemble models model.

Another essential statistic for evaluating classification model performance is the receiver operating characteristic (ROC) curve, which plots true positive rate (TPR), or sensitivity, and false positive rate (FPR), or 1-specificity, at various classification thresholds. Equations (5) and (6) provide the formulas for computing TPR and FPR, respectively:

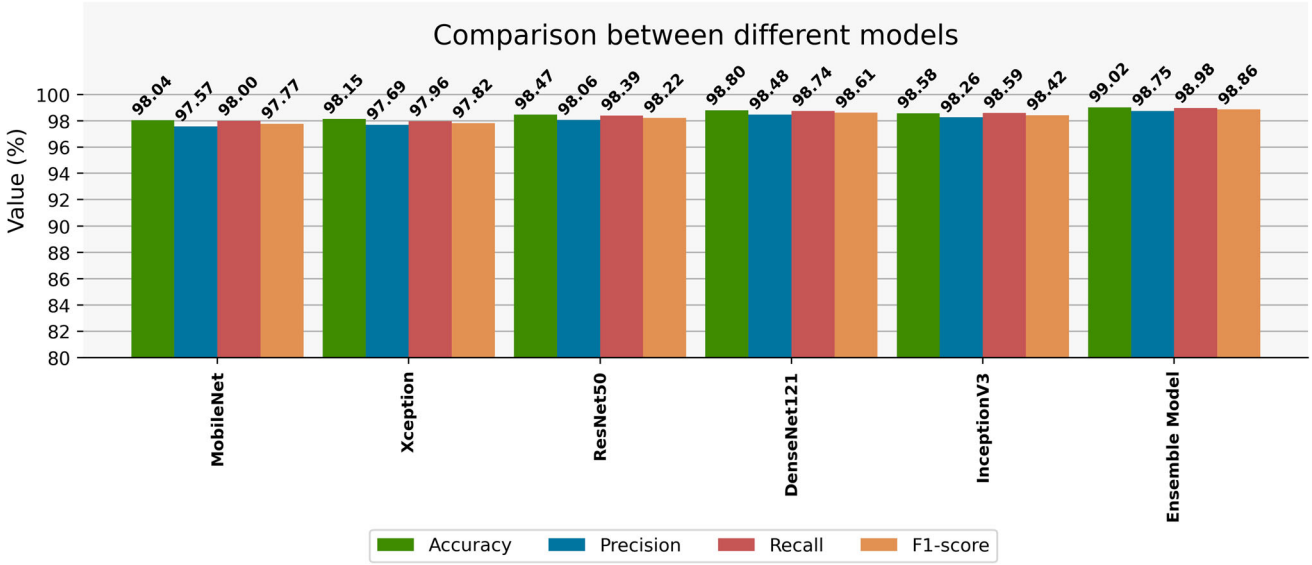

Fig. 6 Evaluation metrics comparison between various pre-trained networks and the ensemble model

$$TPR = \frac{TP}{TP + FN} \tag{5}$$

$$FPR = \frac{FP}{FP + TN} \tag{6}$$

The ROC curve shows how well the model separates between classes. The area under the ROC curve (AUC) is the area underneath the ROC curve in which superior models have an AUC near 1. Figure 8 shows the ROC curves for pre-trained models and the ensemble model for each type of brain tumor in the CE-MRI dataset.

4.2 Evaluation using cross-validation

We used the stratified k-fold cross-validation method for further evaluation, which splits the data into k numbers of folds. At each iteration, the model will be trained on k-1 folds and the remaining fold for validation. A stratified k-fold guarantees that each fold will have the same ratio of samples for each class, whereas a standard k-fold does not. To make the evaluation more challenging, we evaluated the model using stratified k-fold cross-validation with five folds against only the training data for training and validation. In contrast, the test data was used for the final evaluation. Figure 9 shows the division of the data into training, validation, and testing.

The models DenseNet121, InceptionV3, and Ensemble were evaluated for five iterations. The models were trained on different data splits in each iteration, and the classifier weights were adjusted using the same method and hyperparameters. After training, the models were evaluated against the test data, and the evaluation metrics were stored. Each model was trained and validated on 70% of the dataset using different splits, while the remaining 30% was used for the final evaluation. The ensemble model was able to achieve a mean accuracy of 98.11%. Table 6 shows the accuracy of each iteration computed on the test data.

4.3 Grad-CAM results

We tested the explainable Grad-CAM model using unseen samples from the test dataset. Grad-CAM could correctly highlight significant parts of both networks on MR images from three distinct planes (axial, coronal, and sagittal). Figures 10 and 11 show the results of the explainable ensemble model of DenseNet121 and InceptionV3. Figure 12 displays Grad-CAM results on a set of DenseNet121 blocks, and Convolutional layers are sorted as earlier layers, starting from the top left, while later ones are present at the bottom right.

4.4 Computational complexity analysis

The suggested approach relies on transfer learning. It employs pre-trained models that have already adjusted weights and do not need to be retrained or readjusted from scratch as they can detect features of similar tasks, which reduces time and uses fewer hardware resources, as shown in Figs. 13 and 14. Moreover, the classification performance of the proposed ensemble approach comes at the expense of computational complexity compared to a standalone model, which will have fewer parameters and less time to run, as illustrated in Fig. 15. Additionally, the proposed ensemble approach did not enhance the extracted features of the models, as they were trained independently

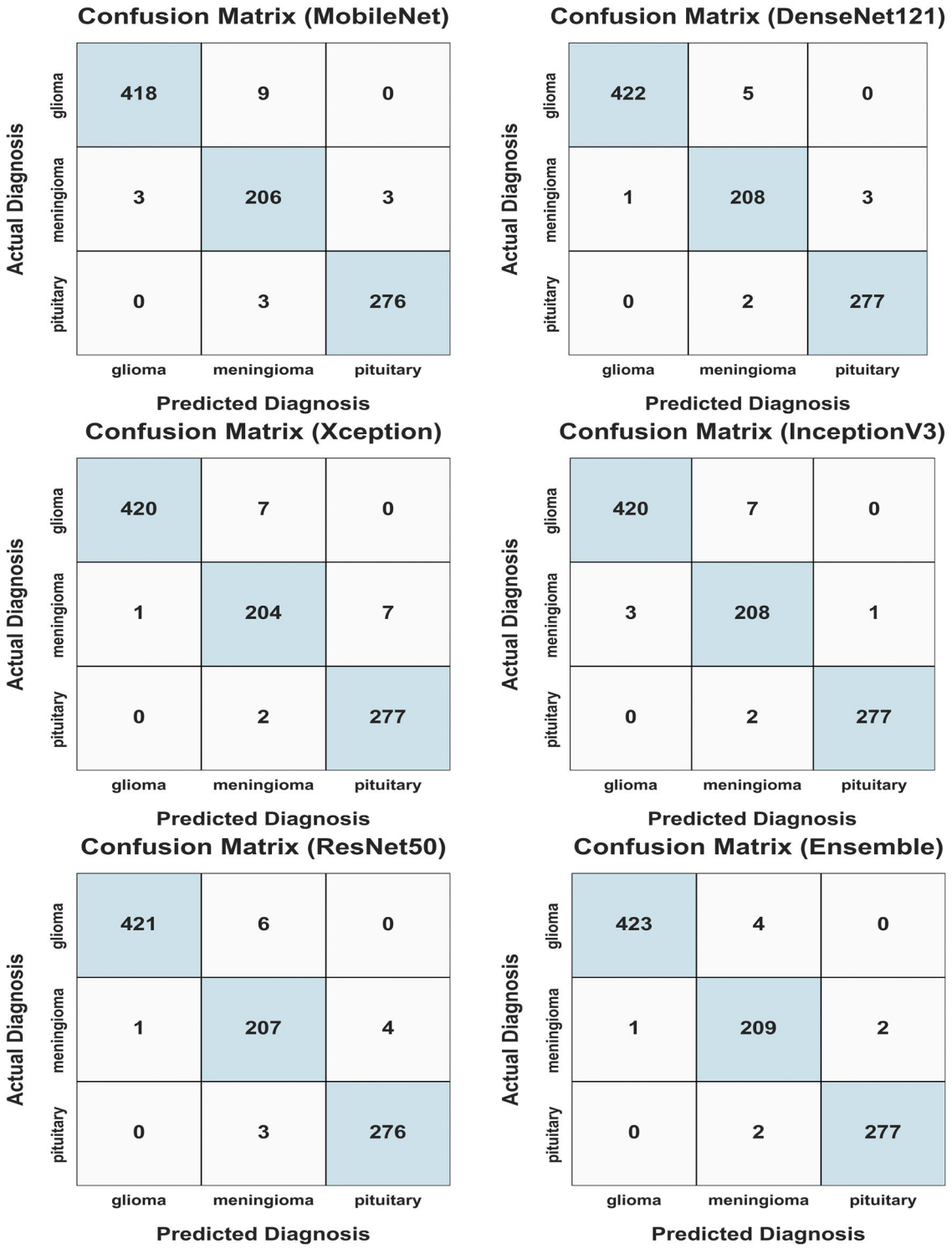

Fig. 7 Confusion matrix of the pre-trained networks and the ensemble model

and had no further interaction during training. Developing a hybrid architecture of CNN and ML algorithms, or visual descriptors, can solve these issues because it will use fewer machine resources and enrich the final feature vector.

4.5 Comparison with current classification methodologies

Previously, researchers employed ML and DL algorithms in their investigations to categorize and detect distinct brain

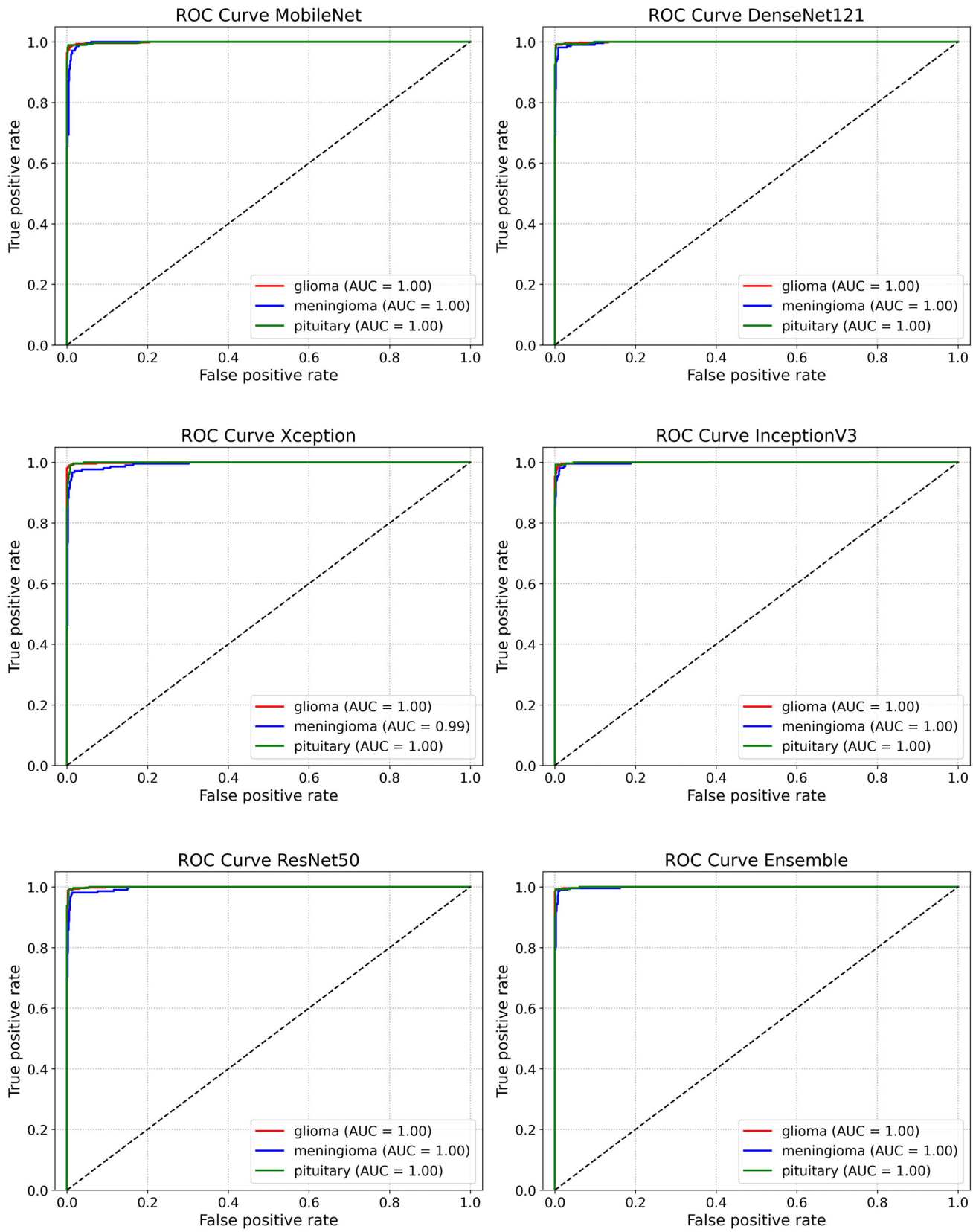

Fig. 8 ROC curves with AUC for pre-trained networks and the ensemble model

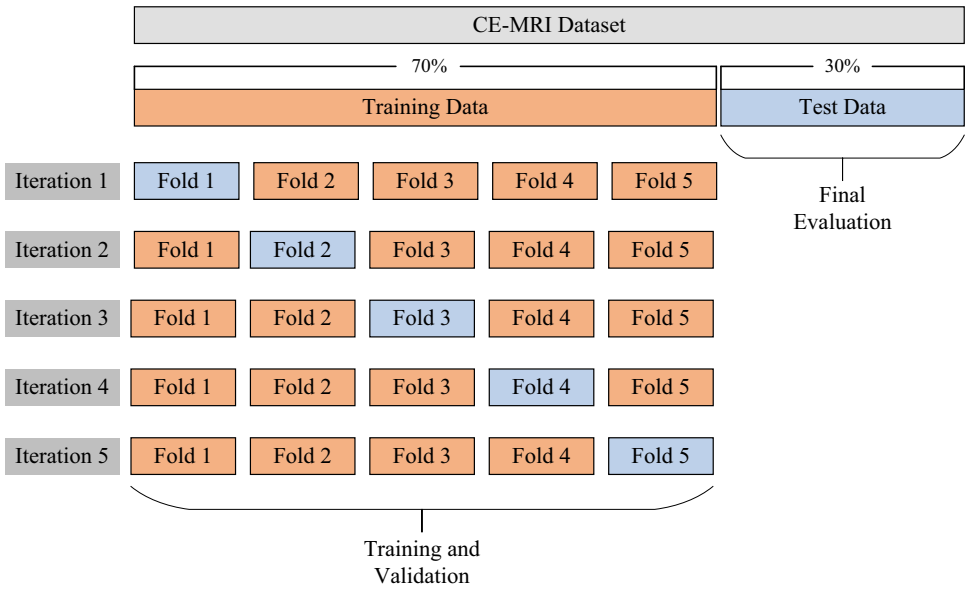

Fig. 9 Dataset distribution of stratified fivefold cross-validation

Table 6 Accuracy for each model using stratified fivefold cross-validation

Model	Accuracy					
	Iteration 1 (%)	Iteration 2 (%)	Iteration 3 (%)	Iteration 4 (%)	Iteration 5 (%)	Mean (%)
DenseNet121	98.04	97.49	98.15	97.71	97.71	97.82
InceptionV3	97.93	97.17	97.93	98.04	97.39	97.69
Ensemble (DenseNet121 + InceptionV3)	98.15	98.15	98.04	98.15	98.04	98.11

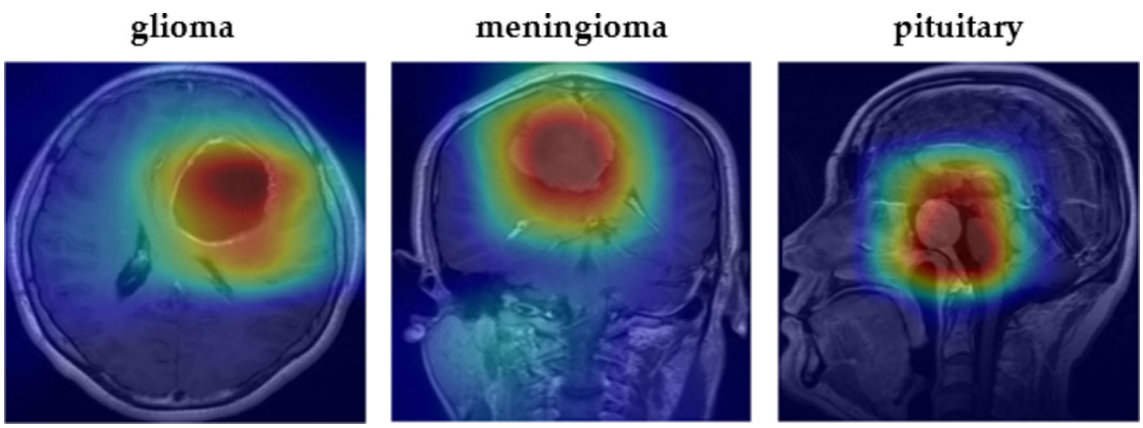

Fig. 10 Grad-CAM results from DenseNet121

tumors on the same dataset. We compared the results of the suggested approach with those from earlier studies in the literature [21, 24, 26, 39–41] that used a similar dataset. Table 7 and Figure 16 compare the most recent methods with the proposed methodology, and it is noticeable that the suggested approach outperformed earlier research, even though it used fewer training instances and more testing instances. Ensemble learning helped to increase accuracy and achieve superior performance by combining weaker models to produce more robust and enhanced models.

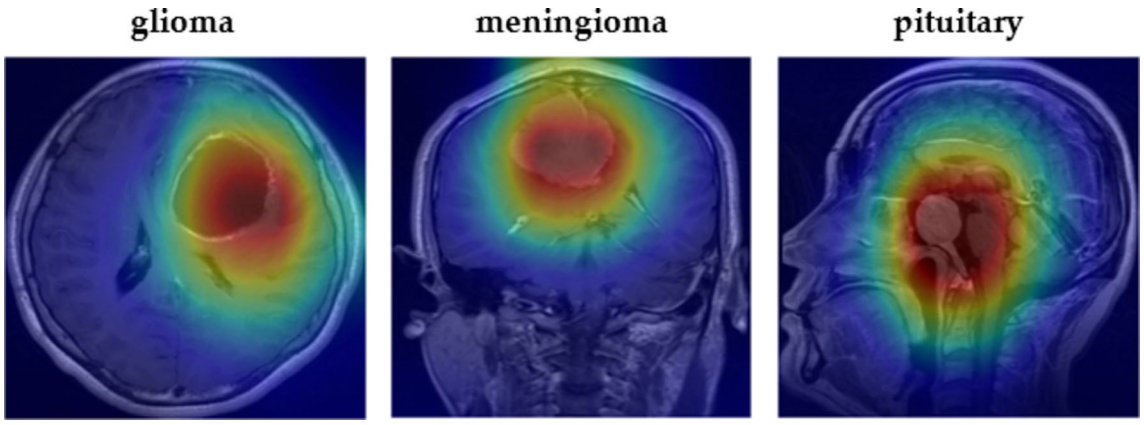

Fig. 11 Grad-CAM results from InceptionV3

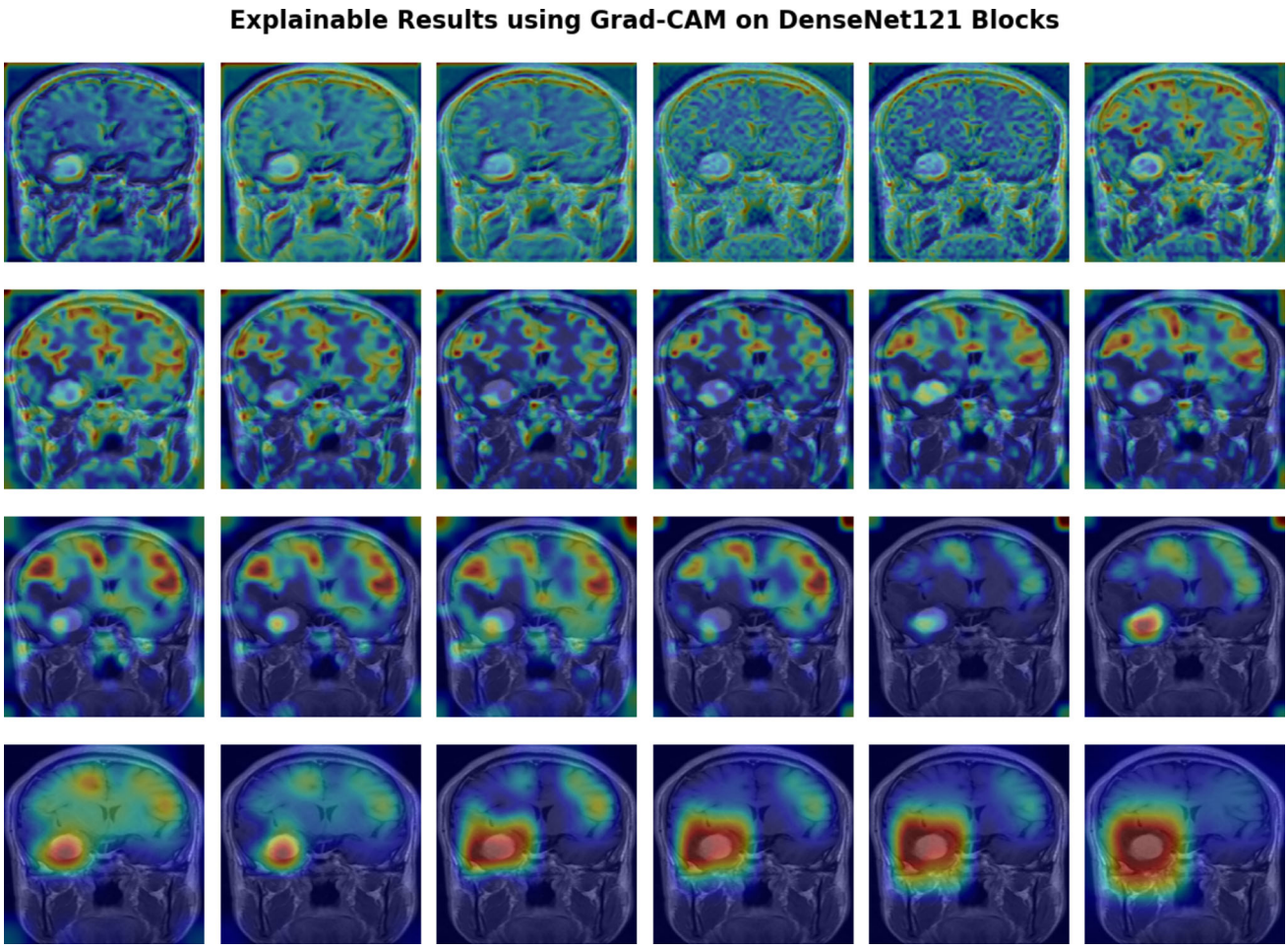

Fig. 12 Grad-CAM results on DenseNet121 Blocks

5 Discussion

This study provides an explainable ensemble model for detecting three brain malignancies (meningioma, glioma, and pituitary) through transfer learning using DenseNet121 and InceptionV3 pre-trained models. Ensemble learning improved overall model performance because it did not rely on individual model predictions or features. The model misclassified nine samples out of 918 and achieved 99.02% accuracy, 98.75% precision, 98.98% recall, and a 98.86% F1 score. The proposed method was compared to prior works that used the same dataset with different train–test

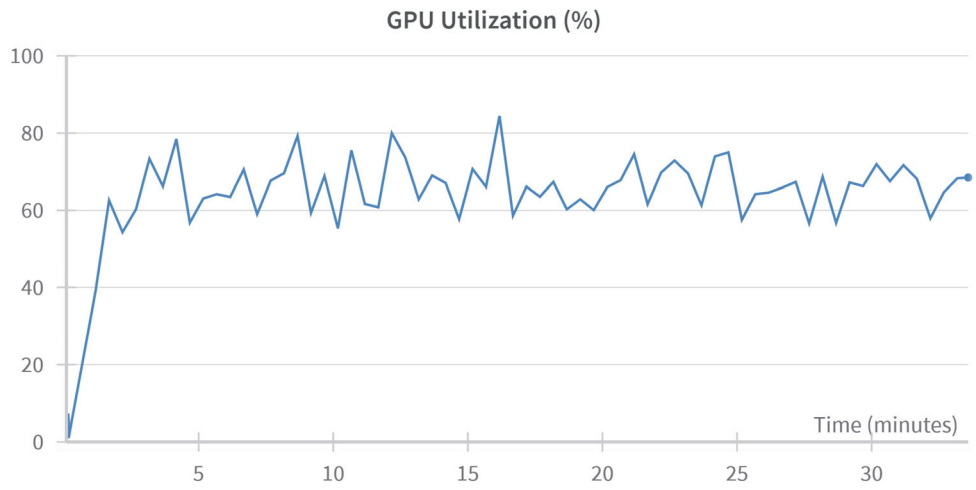

Fig. 13 Resource utilization and training time of DenseNet121

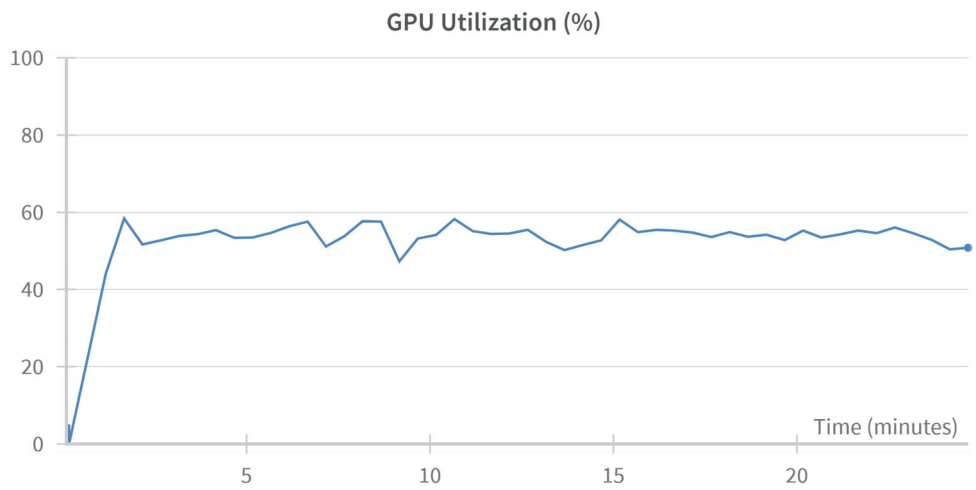

Fig. 14 Resource utilization and training time of InceptionV3

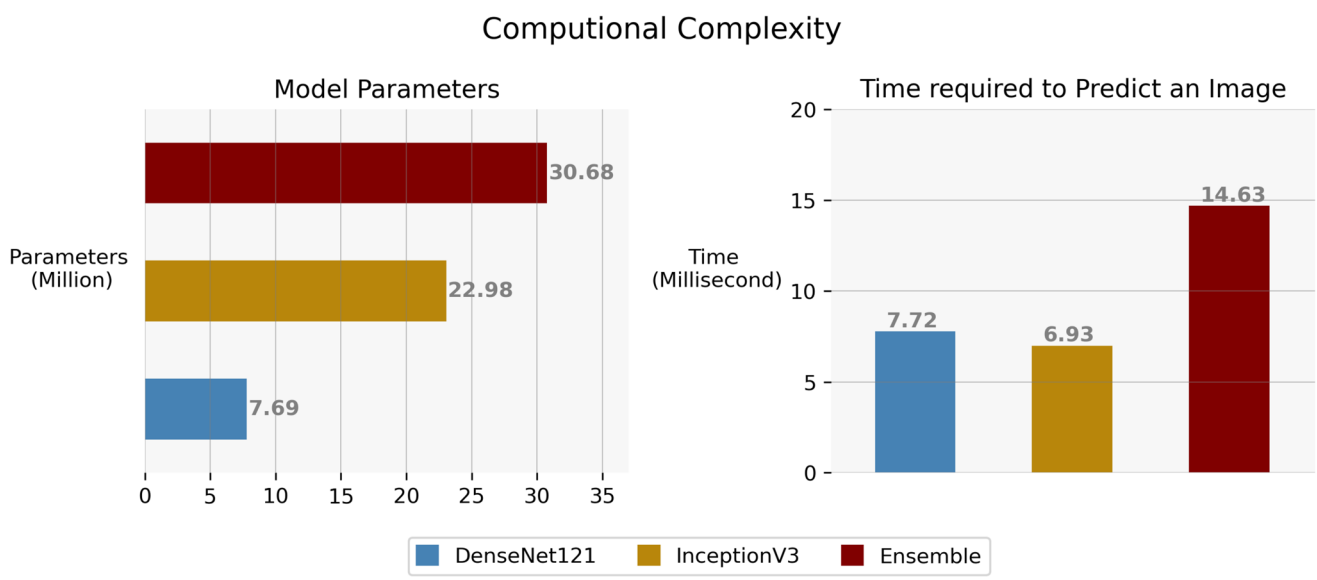

Fig. 15 Computational complexity of the ensemble model compared to DenseNet121 and InceptionV3

Table 7 Comparisons with earlier studies utilized a similar dataset (CE-MRI)

Author	Model	Accuracy (%)	Train–test ratio (%)
Verma et al. (2022) [21]	Transfer learning using DenseNet201	98.22	70–30
Athisayamani et al. (2023) [24]	ResNet-152	98.85	83–17
Kumar et al. (2021) [26]	ResNet-50 and global average pooling	98.00	80–20
Rasheed et al. (2023) [39]	CNN	98.04	80–20
Alhassan et al. (2021) [40]	Swish-based RELU activation function-CNN	98.60	80–20
Malla et al. (2023) [41]	VGG-16 with fine-tuning	98.93	85–15
Proposed method	Ensemble model (DenseNet121 + InceptionV3)	99.02	70–30

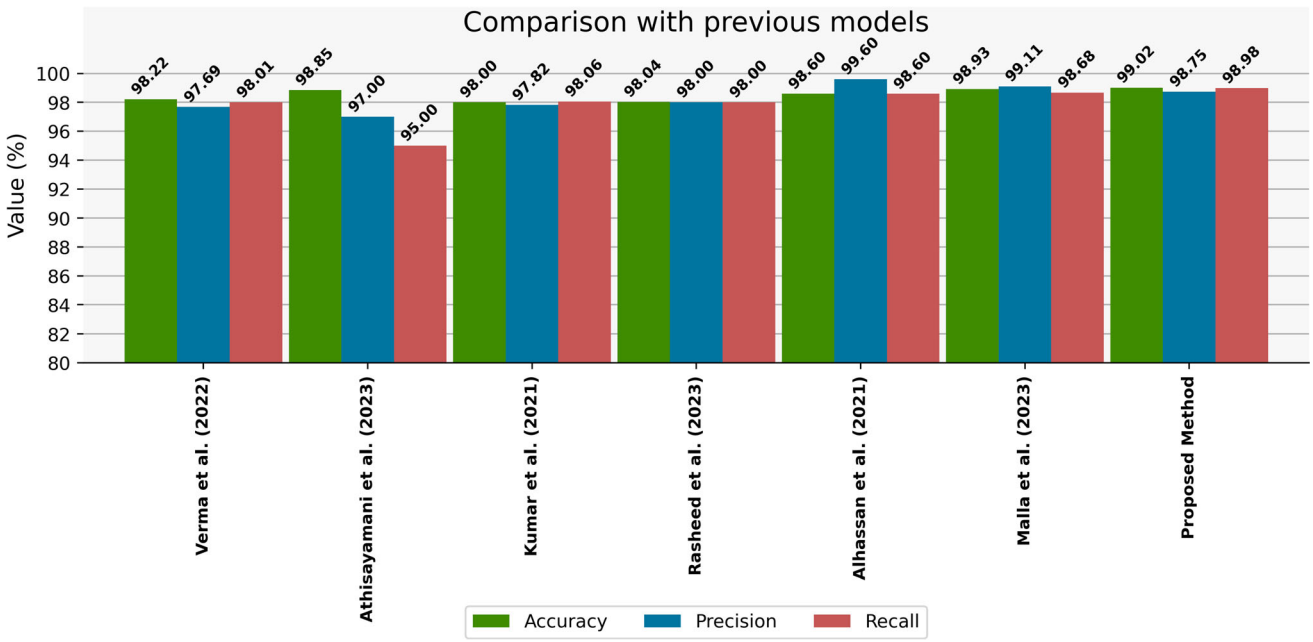

Fig. 16 Evaluation metrics comparison with previous research models

split ratios. As shown in Table 7 and Fig. 16, the performance of the proposed method was superior to previous works, even though it used fewer training samples and more testing instances.

The difference in classification performance between pre-trained networks and the ensemble model is noticeable in the confusion matrix obtained in Fig. 7. DenseNet121 failed to classify 11 samples, while InceptionV3 misclassified 13 samples. The ensemble model could classify two more images than DenseNet121 and four more images than InceptionV3. In Fig. 8, the ROC curves of pre-trained models and the ensemble model show that the ensemble model's ROC curve is closer to one than the ROC curves of other pre-trained models, indicating that it can separate and distinguish between the three brain malignancies better.

Further evaluation was conducted to test the reliability of the model's accuracy. The model was evaluated using stratified 5-fold cross-validation, which resulted in a mean accuracy of 98.11%. Table 6 shows that during the evaluation process, we tested the ensemble model on 30% of the dataset in each iteration, achieving an accuracy of over 98.00% in all iterations. Additionally, it outperformed the pre-trained models in all iterations except one, demonstrating its superior accuracy.

Grad-CAM was used in the proposed model to explain the results of the DenseNet121 and InceptionV3 pre-trained models, which can be seen in Figs. 10 and 11. Both models were able to highlight affected areas that are broadly similar but not entirely the same, as each model extracts features differently. Grad-CAM reflected this by obtaining significant parts that dominate the predicted class. While the earlier layers in Fig. 12 show low-level features, the model gradually captures more complex patterns in the later layers that participate most in the classification phase.

The explainable model enhances confidence and facilitates the diagnostic process.

6 Conclusions

This research article aimed to classify different brain tumors effectively using ensemble learning based on transfer learning with explainable AI. Preprocessing was done on the dataset by extracting ROI through applying a threshold, cropping, and resizing to remove the unnecessary black parts of the image. Two pre-trained models (DenseNet121 and InceptionV3) were used with a new classifier to adopt the recent task. The dropout layer and L2 regularization were used to overcome the overfitting problem. Also, data augmentation was applied to make the model more robust, expand the size of training instances, and overcome imbalanced classes.

The model was evaluated against a publicly available dataset using two methods: Train–test split and stratified k-fold cross-validation to ensure that the model is reliable. An explainable model was adopted to highlight the significant parts, and testing was done on MR images obtained on different planes. The ensemble model solves the shortage in prior works, as combined models can perform better in classification than single models by extracting predictions or features from more than one model. The explainable model can highlight the affected areas in MR images, increasing the trust level of the model that neurologists and surgeons can use.

In the future, we seek to develop a more efficient and effective hybrid approach using DL and ML techniques that can reduce the number of parameters, optimize machine resources, and enhance the extracted feature vector.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article contains no studies with human participants performed by any authors.

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